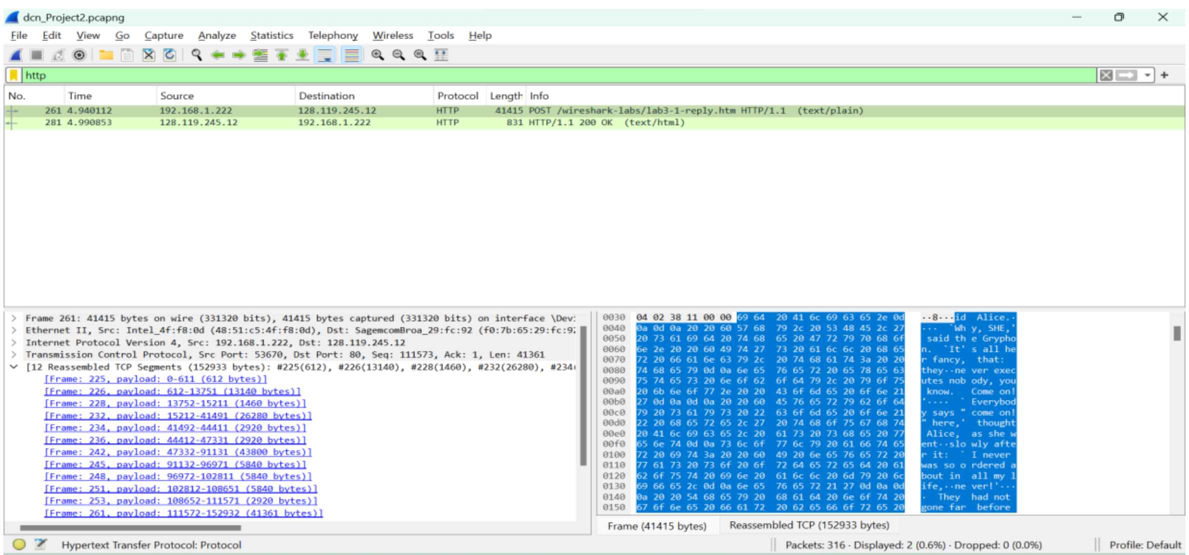
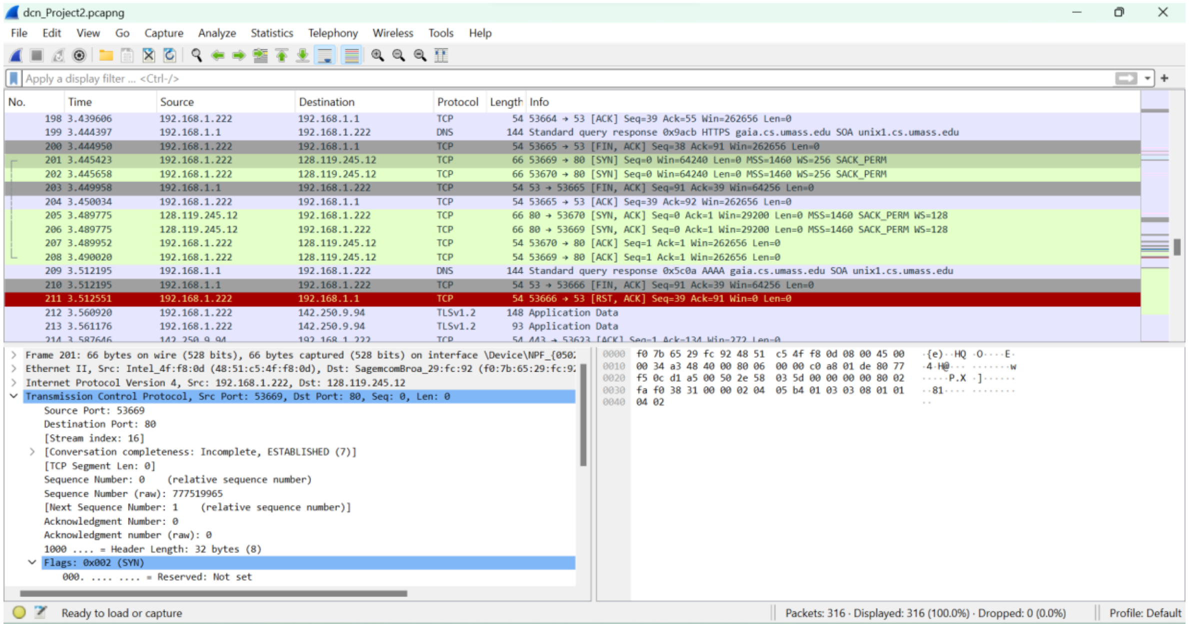
Data Communications and Networking
Project 2
Wireshark Lab: TCP

<u>Answer the following questions, from your captured trace.</u>

1. What is the IP address and TCP port number used by the client computer (source) that is transferring the alice.txt file to gaia.cs.umass.edu? (Hint: To answer this question, it's probably easiest to select an HTTP message and explore the details of the TCP packet used to carry this HTTP message, using the "details of the selected packet header window").

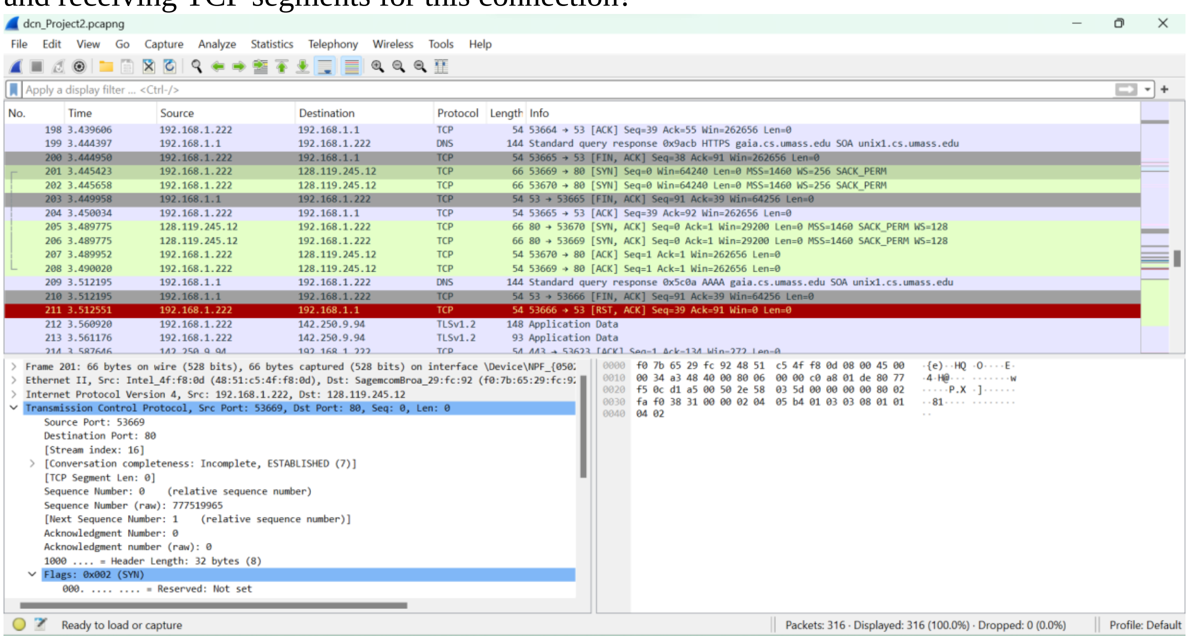
   <u>Answer:</u>





The IP address and TCP port number used by the client computer (source) that is transferring the alice.txt file to gaia.cs.umass.edu are **192.168.1.222** and TCP port number used by the client computer is **53669.**

2. What is the IP address of gaia.cs.umass.edu? On what port number is it sending and receiving TCP segments for this connection?

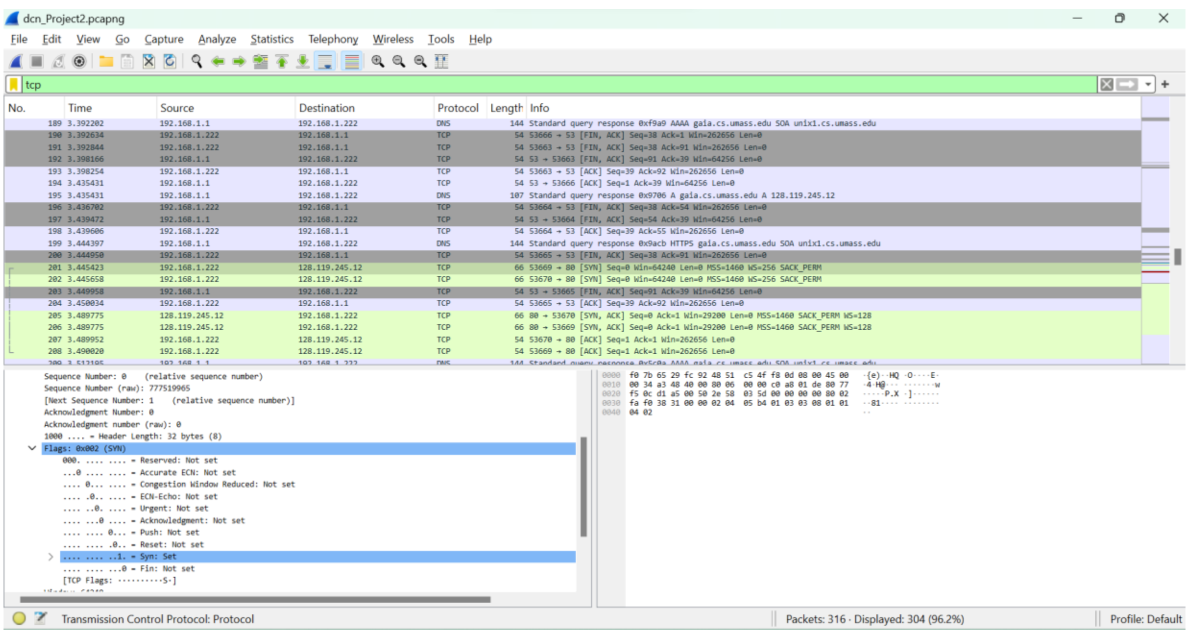


The IP address of gaia.cs.umass.edu is **128.119.245.12** and the it is sending and receiving TCP segments on port number **80**.

1. TCP Basics

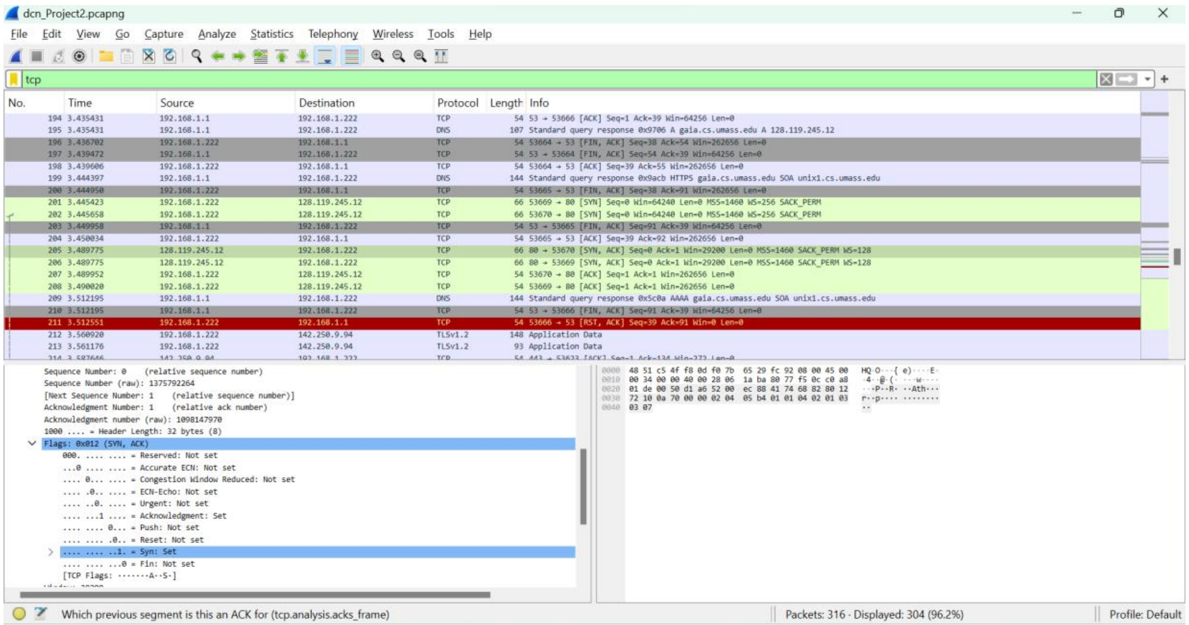
Answer the following questions for the TCP segments:

3. What is the *sequence number* of the TCP SYN segment that is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu? (Note: this is the "raw" sequence number carried in the TCP segment itself; it is *NOT* the packet # in the "No." column in the Wireshark window. Remember there is no such thing as a "packet number" in TCP or UDP; as you know, there *are* sequence numbers in TCP and that's what we're after here. Also note that this is not the relative sequence number with respect to the starting sequence number of this TCP session.). What is it in this TCP segment that identifies the segment as a SYN segment?
   ANSWER:



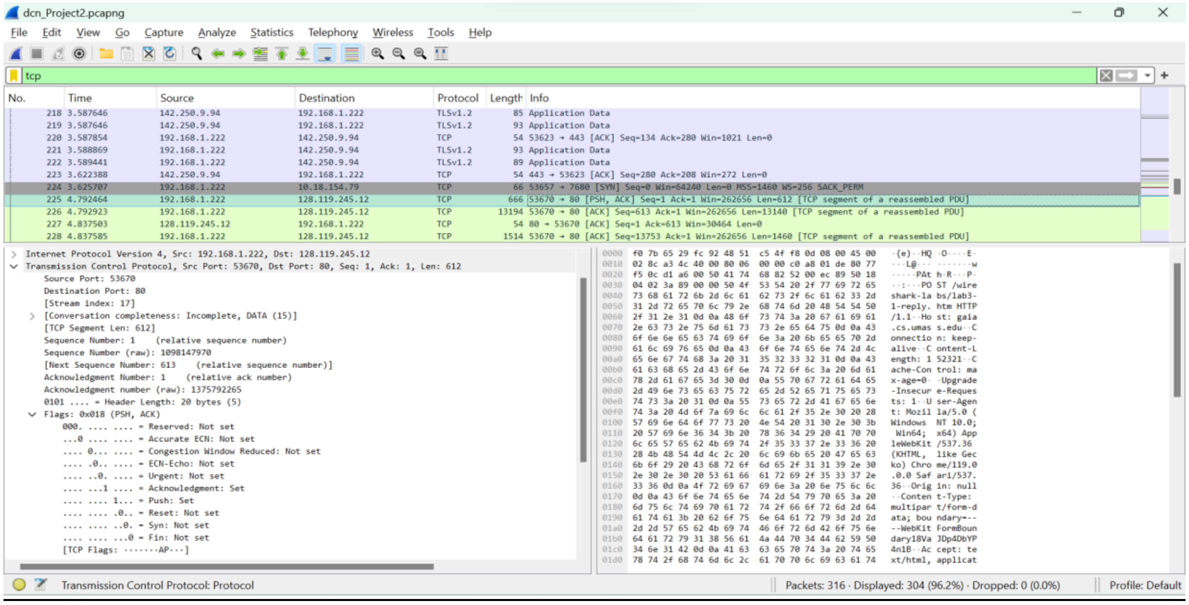
The ***sequence number*** of the **TCP SYN segment** that is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu is **0** in the trace. The SYN flag bit is **set** which identifies the TCP segment as a SYN segment.
Sequence Number(raw): 777519965

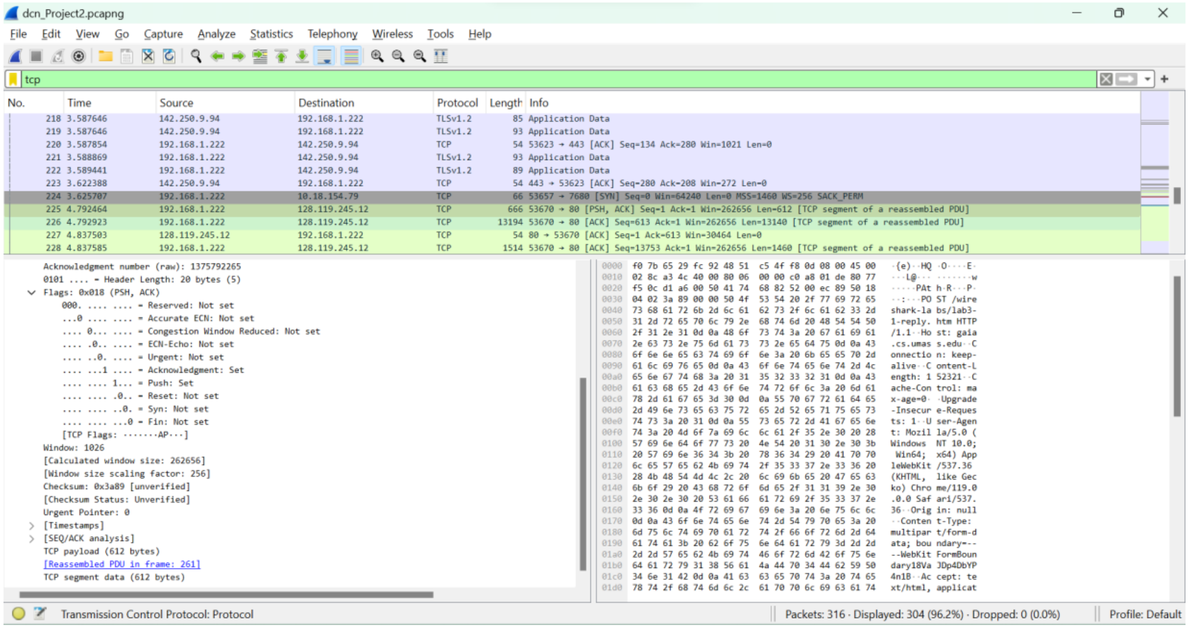
4. What is the *sequence number* of the SYNACK segment sent by gaia.cs.umass.edu to the client computer in reply to the SYN? What is it in the segment that identifies the segment as a SYNACK segment? What is the value of the Acknowledgement field in the SYNACK segment? How did gaia.cs.umass.edu determine that value?
   ANSWER:



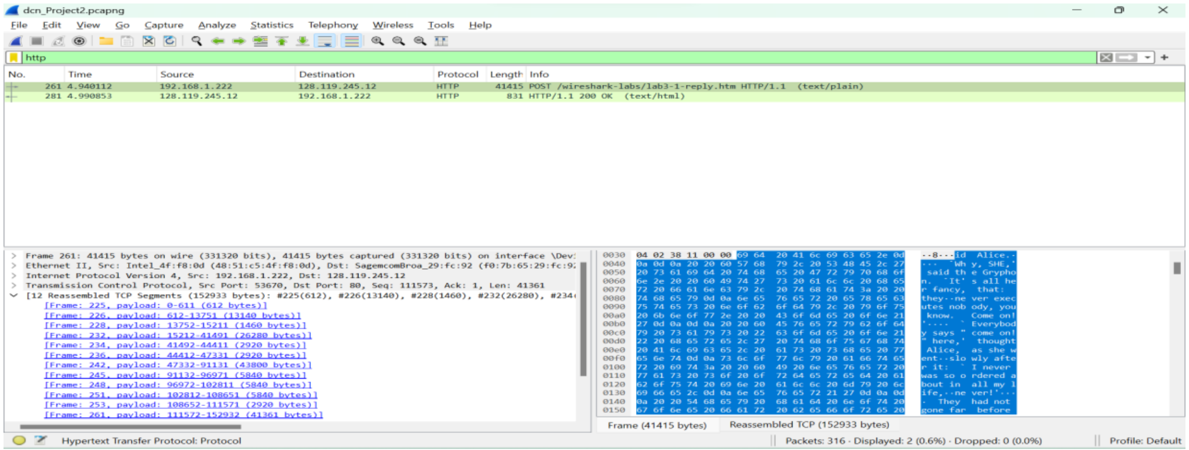
The ***sequence number* of the SYNACK segment** that is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu is **0** in the trace.
The **SYN and ACK flag bits are set** which identifies the segment as a SYNACK segment.
The **Acknowledgement number** in the SYNACK segment is **1.**
gaia.cs.umass.edu determined this value by **adding 1 to the initial sequence number of SYN segment 0.**
Sequence Number(raw): 1375792264
Acknowledgement Number(raw): 1098147970

5. What is the sequence number of the TCP segment containing the header of the HTTP POST command? Note that in order to find the POST message header, you'll need to dig into the packet content field at the bottom of the Wireshark window, *looking for a segment with the ASCII text "POST" within its DATA field*. How many bytes of data are contained in the payload (data) field of this TCP segment? Did all of the data in the transferred file alice.txt fit into this single segment?
<u>ANSWER:</u>



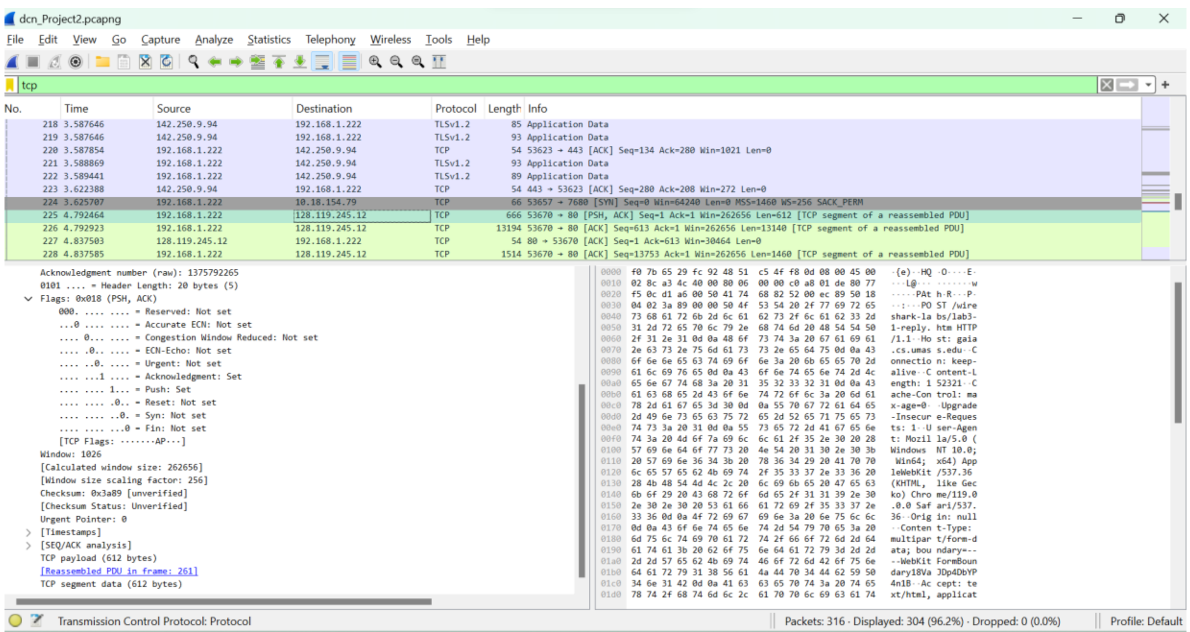


The sequence number of the TCP segment containing the header of the HTTP POST command is **1**.
TCP payload: **612 Bytes**

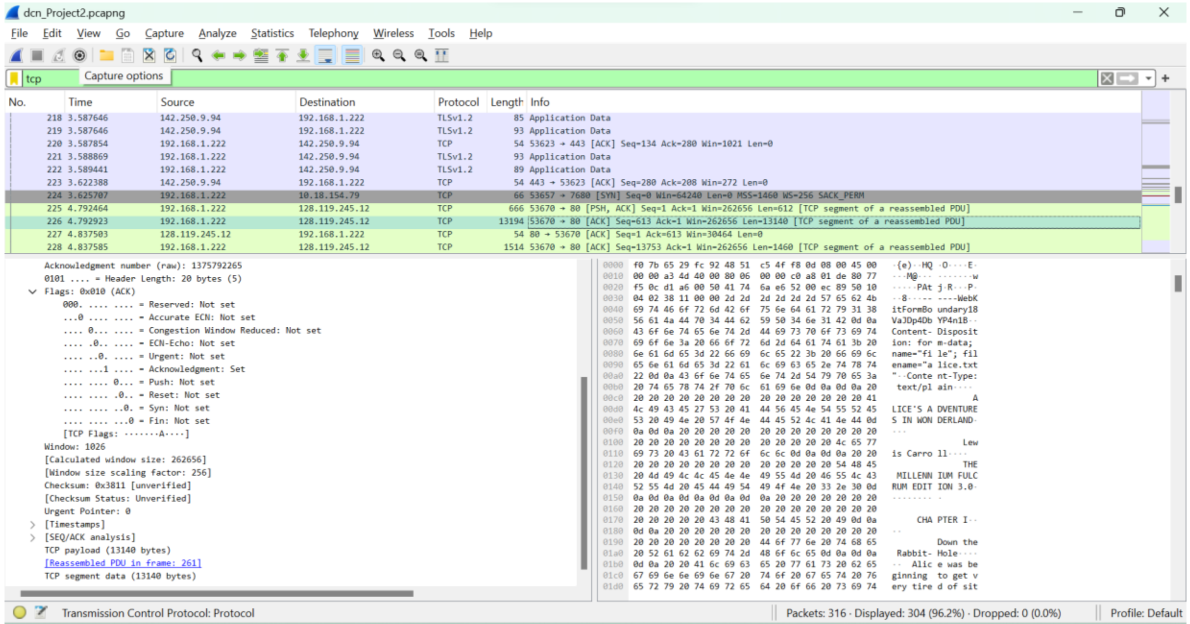


No, all of the data in the transferred file alice.txt did not fit into this single segment.

6. Consider the TCP segment containing the HTTP "POST" as the first segment in the data transfer part of the TCP connection.
   - At what time was the first segment (the one containing the HTTP POST) in the data-transfer part of the TCP connection sent?
     <u>ANSWER:</u>
     The time, the first segment sent in data transfer part of TCP connection is **4.792464.**



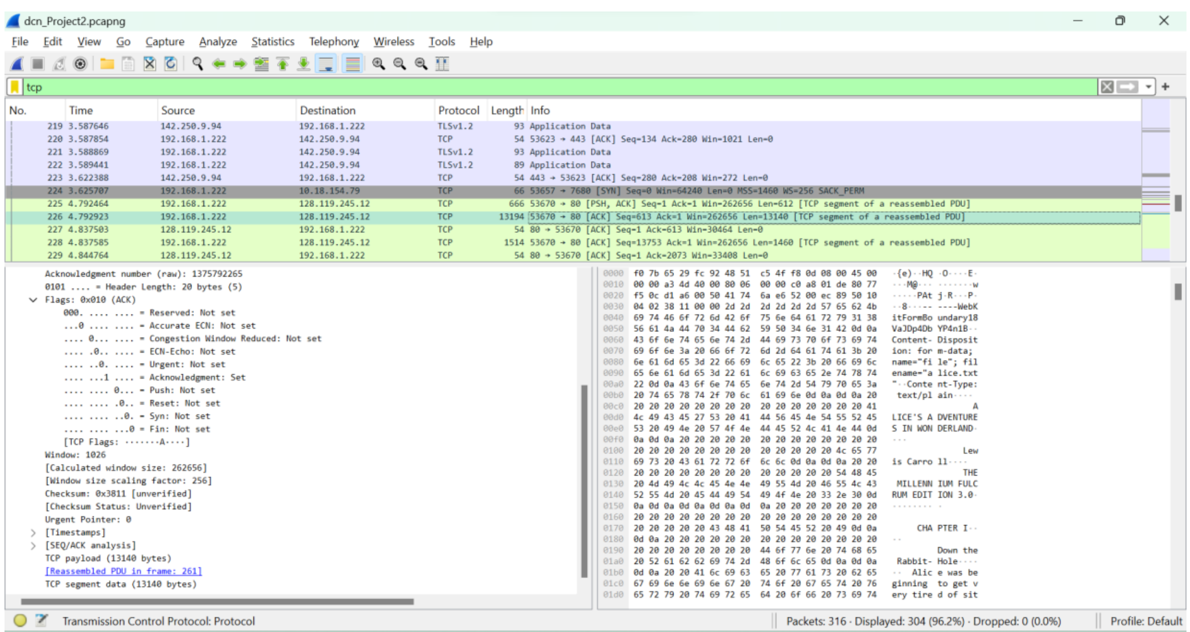
   - At what time was the ACK for this first data-containing segment received?
     <u>ANSWER:</u>
     The time, the ACK for this first data-containing segment received was **4.792923**.



- What is the RTT for this first data-containing segment?
  <u>ANSWER:</u>

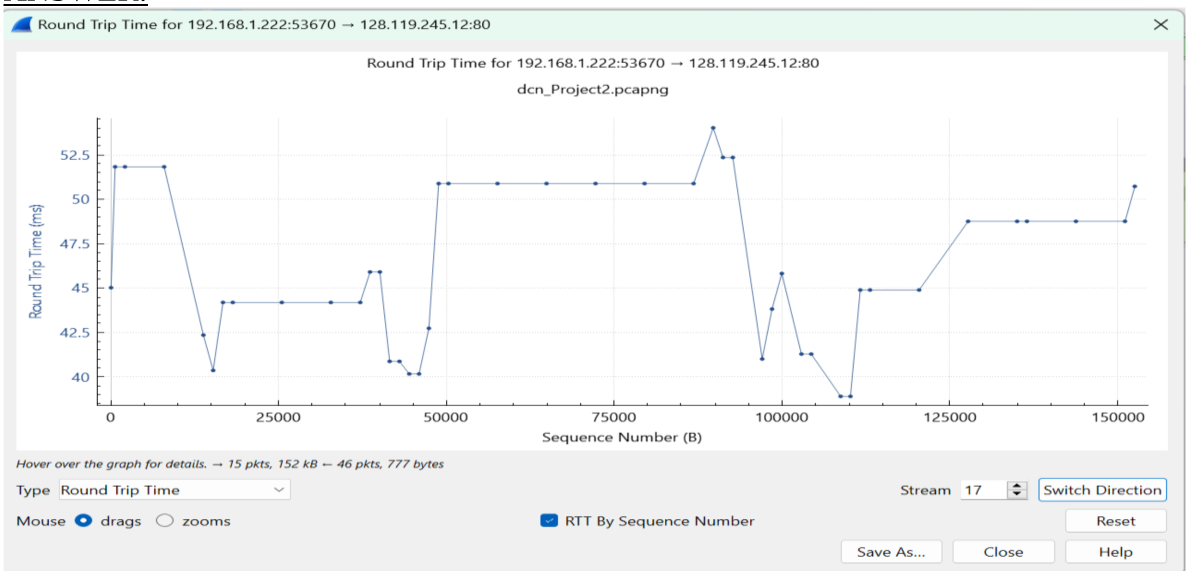
  The RTT for this first data-containing segment is **4.792923[ACK received time]- 4.792464[1st data segment transferred time]** = **0.000459**.

- What is the RTT value the second data-carrying TCP segment and its ACK?
  <u>ANSWER:</u>

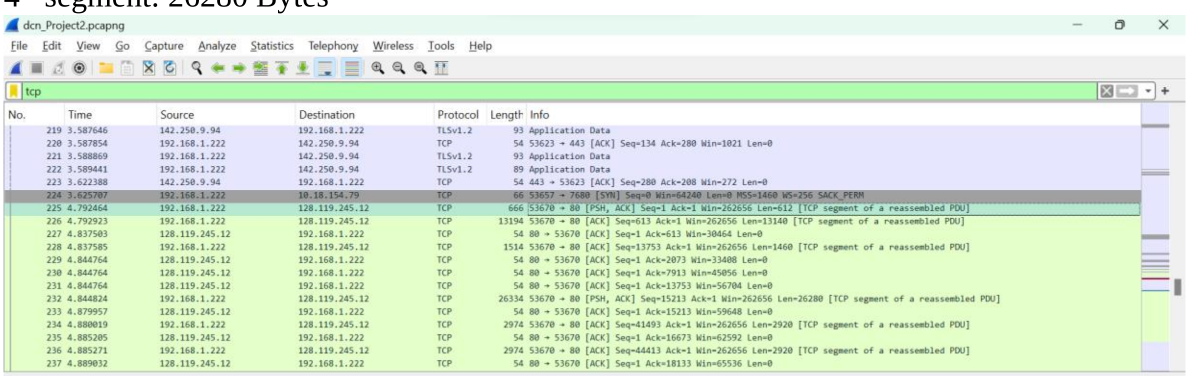


The RTT for the second data-carrying TCP segment is 4.837503 – 4.792923 = **0.04458.**

- Wireshark has a nice feature that allows you to plot the RTT for each of the TCP segments sent. Select a TCP segment in the “listing of captured packets” window that is being sent from the client to the gaia.cs.umass.edu server. Then select: *Statistics->TCP Stream Graph->Round Trip Time Graph.* Plot the RTT and upload a screenshot.
  ANSWER:

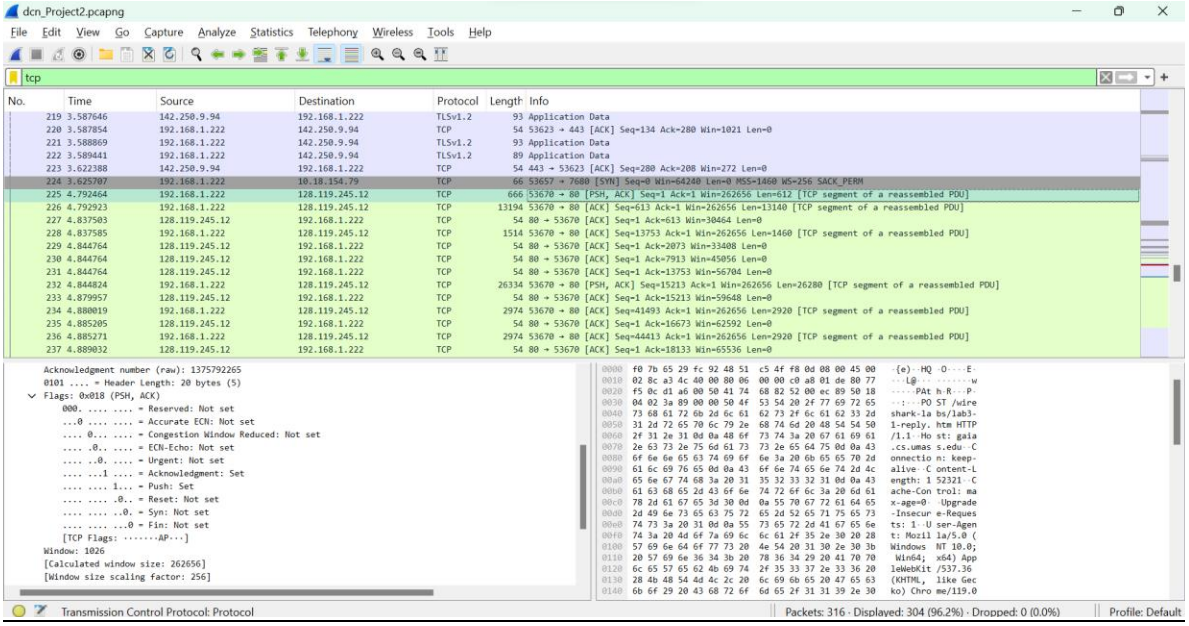


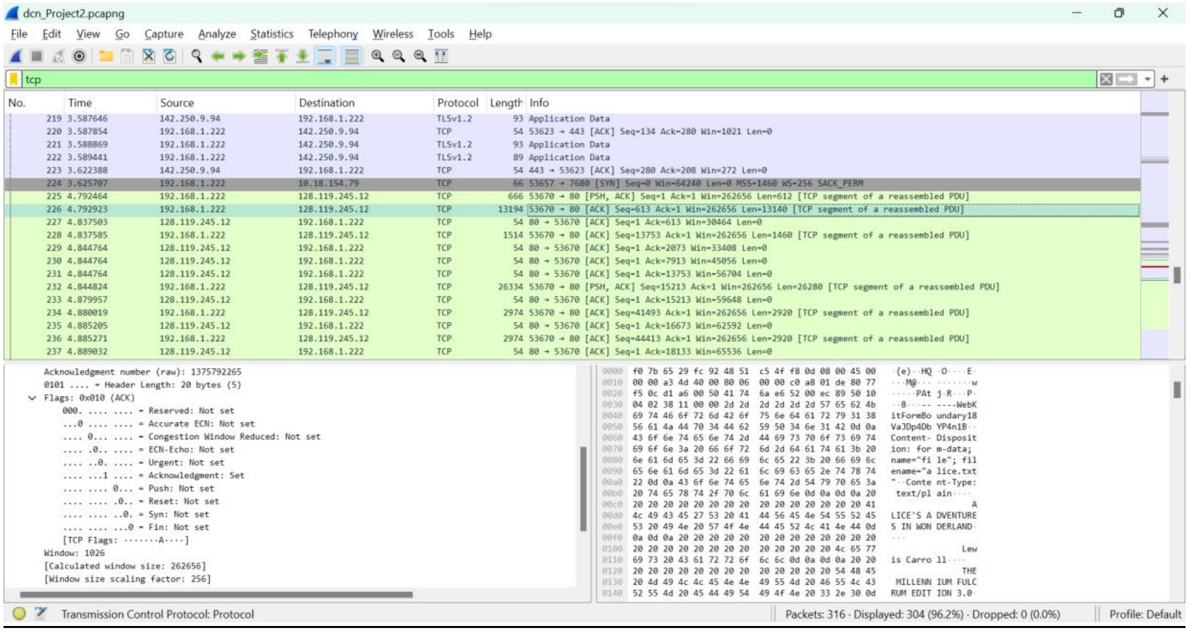
7. What is the length (header plus payload) of each of the first four data-carrying TCP segments?
   ANSWER:
   The length (header plus payload) of
   1st segment: 612 Bytes
   2nd segment: 13140 Bytes
   3rd segment: 1460 Bytes
   4th segment: 26280 Bytes

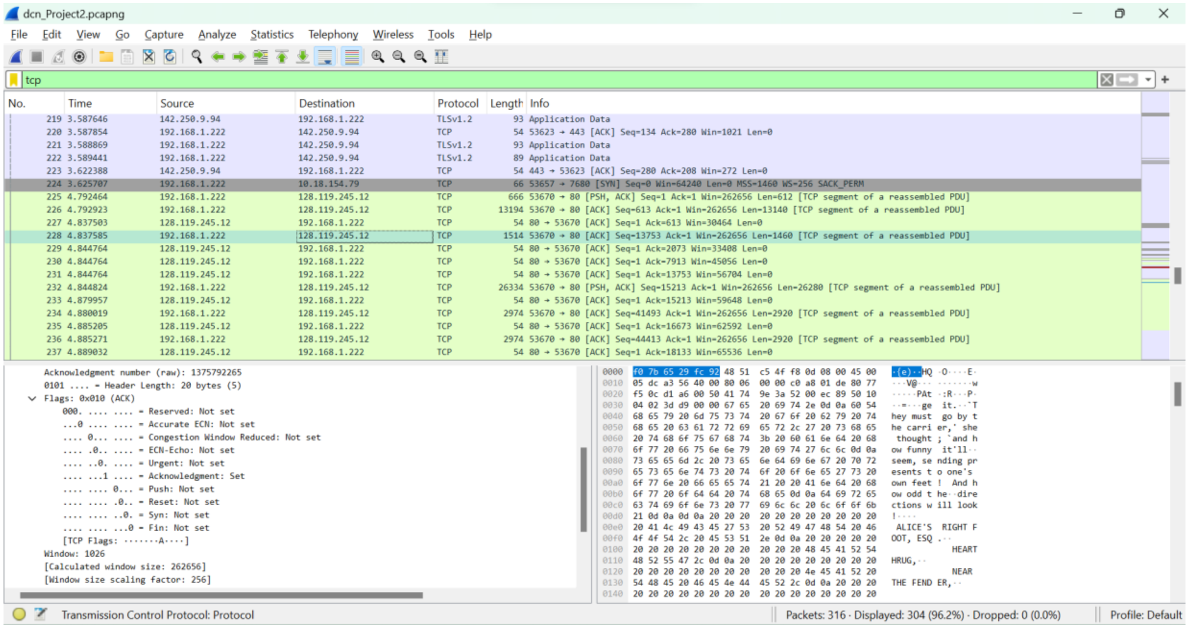


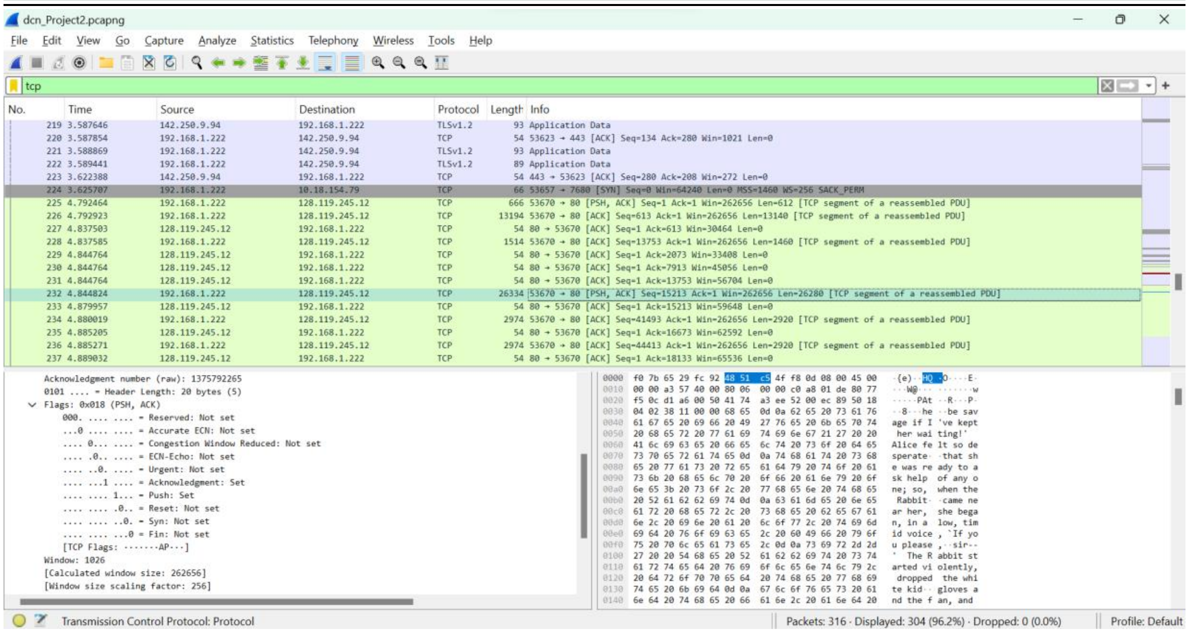
8. What is the minimum amount of available buffer space advertised to the client by gaia.cs.umass.edu among these first four data-carrying TCP segments? Does the lack of receiver buffer space ever throttle the sender for these first four data carrying segments?

ANSWER:









The minimum amount of available buffer space advertised to the client by gaia.cs.umass.edu among these first four data-carrying TCP segments is 1026.

No, the lack of receiver buffer space doesn't throttle the sender for these first four data carrying segments

9. Are there any retransmitted segments in the trace file? What did you check for (in the trace) in order to answer this question?
<u>ANSWER:</u>

No, there are no retransmitted segments in the trace field. The sequence number is increasing throughout the trace.

10. How much data does the receiver typically acknowledge in an ACK among the first ten data-carrying segments sent from the client to gaia.cs.umass.edu? Can you identify cases where the receiver is ACKing every other received segment among these first ten data-carrying segments?

<u>ANSWER:</u>

The receiver typically acknowledge in an ACK among the first ten data-carrying segments sent from the client to gaia.cs.umass.edu has

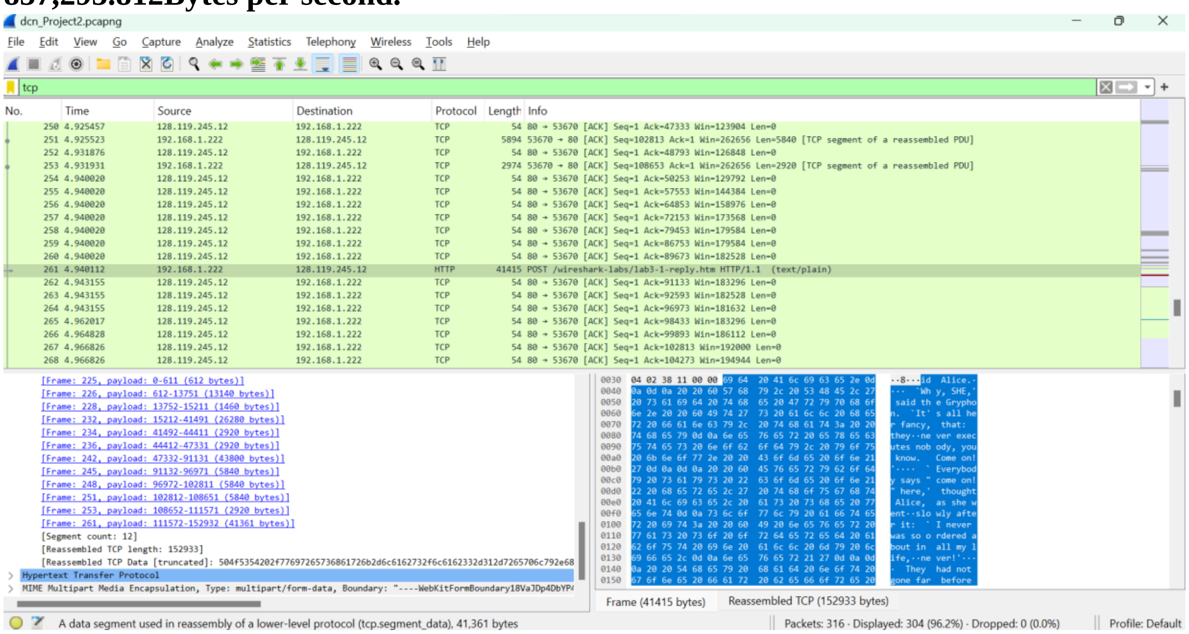
11. What is the throughput (bytes transferred per unit time) for the TCP connection? Explain how you calculated this value.

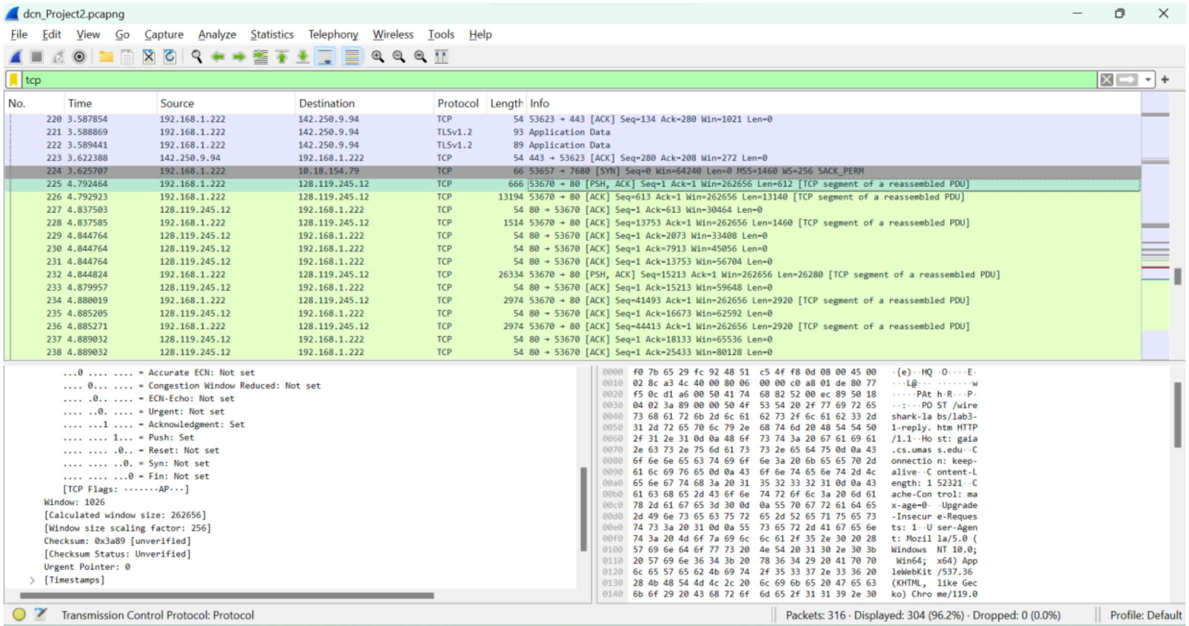
<u>ANSWER:</u>

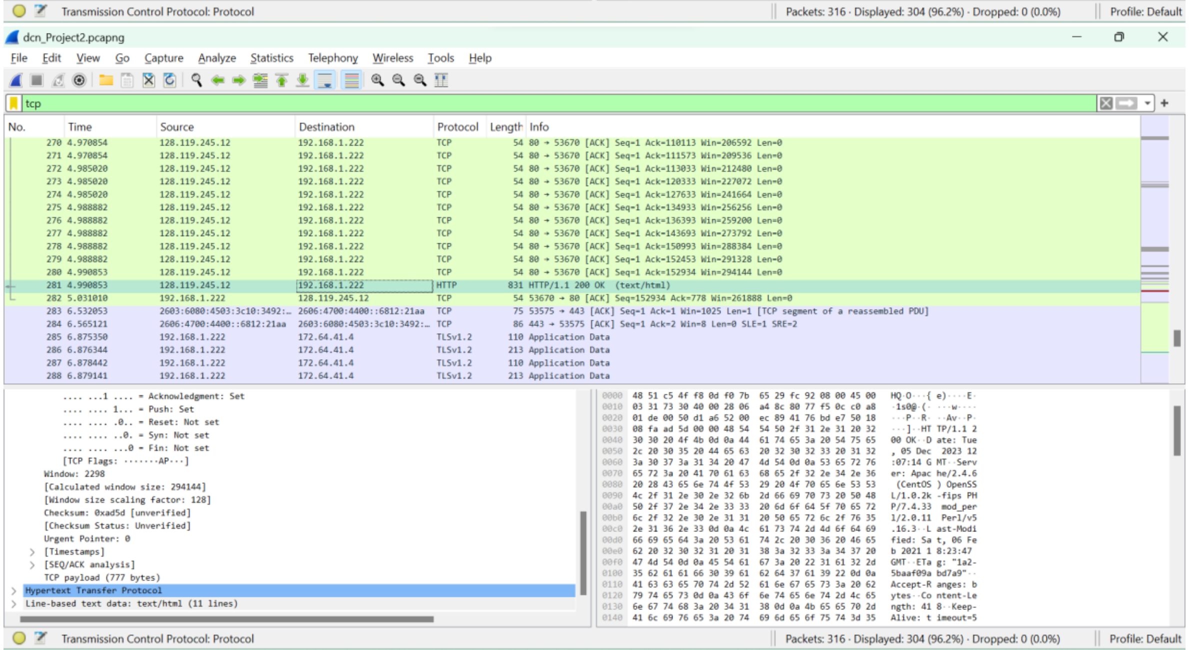
The total data of alice.txt is 152933Bytes. The transmission time is 4.970854(last ACK) – 4.792464 (1^{st} TCP seg) = 0.17839.

The throughput = total data of file/ transmission time = 152933/0.17839=

857,295.812Bytes per second.

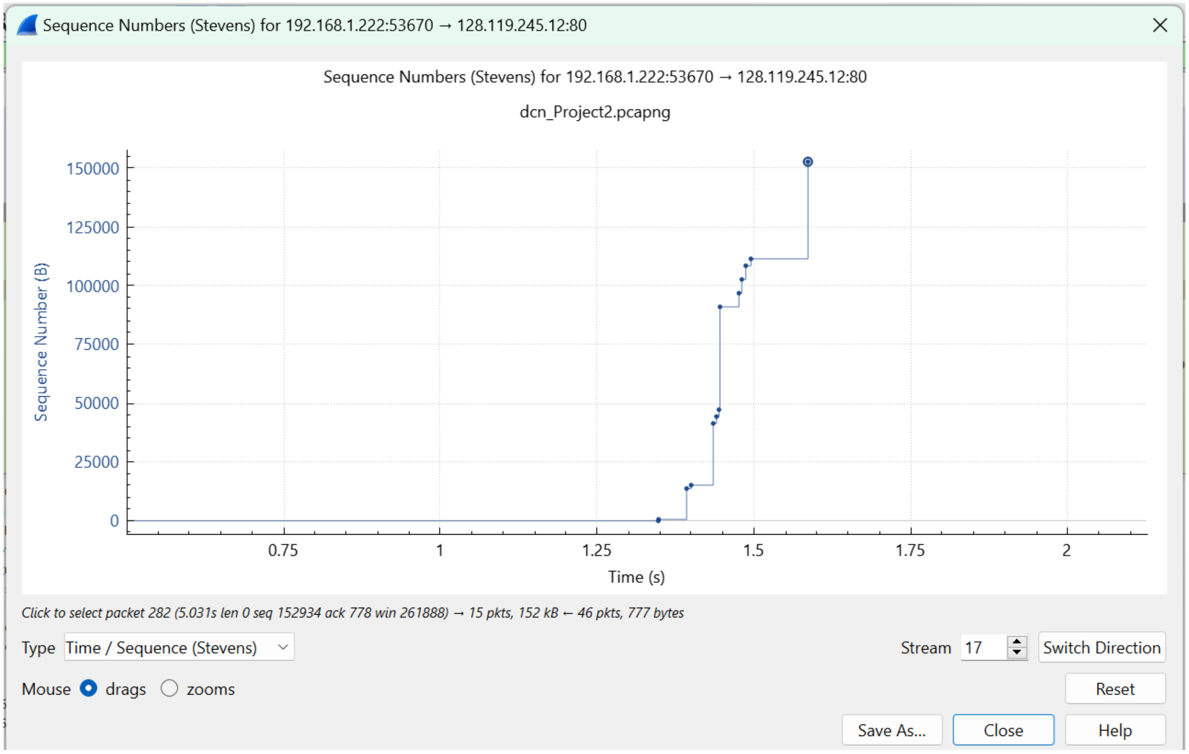






2. TCP congestion control in action

12. Use the *Time-Sequence-Graph(Stevens)* plotting tool to view the sequence number versus time plot of segments being sent from the client to the gaia.cs.umass.edu server. Consider the different “fleets” of packets sent in your graph. Comment on whether this looks as if TCP is in its slow start phase, congestion avoidance phase or some other phase. Figure 6 shows a slightly different view of this data.
    <u>ANSWER:</u>



TCP is in slow start phase and going to congestion avoidance.

13. These “fleets” of segments appear to have some periodicity. What can you say about the period?

<u>ANSWER:</u>

Periodicity is there.

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